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CSCI 332

4/23/23

Software Ownership: Can you really have it all?

Can you have your cake and eat it too? At first glance you may wonder what this long popular saying has to do with the fierce debate over the intellectual property rights that deal with software. However, it may actually be a good way of framing this hot debate, as it all comes down to whether or not ownership of software is all inclusive or rather limited and definable, and if it is how do you go about defining it. This debate at its heart is a matter of ethics and the best way to go about discerning the right answer is to turn to a code of ethics. The Institute of Electrical and Electronics Engineers (IEEE) has a well-defined code of ethics that many, if not most, people can agree lay a good foundation for professionalism and ethics in the field. Using this code of ethics to discuss the intellectual property rights of software can lead to the conclusion that this ownership is not all or nothing, it is much more nuanced, which is necessary for the protection and betterment of both its creators and users.

Before determining how to go about determining the rules for the rights of software it must be determined whether or not it is even necessary in the first place. The argument could be made that contributions to the understanding and furthering of society should not be owned, after all limiting access to important knowledge can slow down or even prevent progress. The IEEE even states that you should “improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems,” and this could be done by not allowing all software and all access to it to be

privately owned and instead allowing anyone or group to have access to it and how it works. (IEEE, 2020) This may seem like a great thing, after all access is often a large barrier to learning. However, another section of the IEEE code of ethics shows that this is dangerous, and ownership is the best course of action but must be defined as it can be taken too far. Software is part of everyday life and as such it can be complicated and even dangerous. The IEEE code of ethics referenced earlier also states that you are “to hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, [and] to protect the privacy of others.” With this taken into consideration you could not in good faith allow everyone access to every part of all software, this could and would put lives at risk and have the potential to ruin livelihoods. Software can be used to further good or bad and it all comes down to whose hands it is in, allowing everyone to have all parts of it whether for a price or for free could have devastating effects.

Since it has been established that it would be detrimental for there to be no ownership rights of software, the next step is to determine the guidelines for ownership. The intellectual property rights are a delicate balancing of giving the user freedom to use the software and protecting the owner of the software rights. Though it comes down to personal opinion and conviction I take the stance that a person cannot own math. This is an important thing to note going forward because all software leads back to math on some level. What I believe can be owned is the application that does the math or some specific way in which the software incorporates the math. I would compare it to the copyright of classical music, the notes and scores themselves are free to use for the public however, the renditions, the actually performances belong to those who perform them. Similarly, it is important for the furtherance of technology that we continue to improve and create new and better software. I believe that open

and closed source code is a good example of this balance. To a degree both of these allow for some ownership, however, to fit the need they are designed for some allow more usage and control of how the software is handled. It all comes down to content and purpose. We cannot allow a person to own the very heart of software, the math of it all, but we must also allow others to make a living and to create wonderful things that is their own to distribute and make income off of as they see fit.

In the end it comes down to one thing, how you interpret the ethical implications that a question this big raises. Not everyone would agree on the IEEE ethics standards used as the basis in this paper, in fact there will never be one code of ethics that everyone will agree on and as such this debate will never truly be settled. However, I personally stand in agreement with the IEEE standards and believe that it is acceptable if not beneficial to interpret them in such a way that it is clear ownership is necessary, though the balance is delicate, and that the accessibility of the heart of it all must remain free for everyone if we are to ensure the security of the future and livelihoods.

Works Cited

“IEEE Code of Ethics.” *IEEE*, Institute of Electrical and Electronics Engineers, June 2020,
<https://www.ieee.org/about/corporate/governance/p7-8.html>.